

Skills Summary

Unit Rates with Whole Numbers

Use this reference while completing your worksheet or when studying.

Finding a Unit Rate: Divide Down to One

Rule: unit rate = total amount $\div$ number of units.

The word **per** names the unit you divide by: dollars *per book* means divide by books; miles *per hour* means divide by hours.

Example: 6 workbooks cost 42 dollars. Find the price per workbook.

Step 1. “Per workbook” says to divide the total cost by the number of workbooks.

Step 2. $42 \div 6 = 7$, so the unit price is **7** dollars per workbook.

Try it: 8 granola bars cost 48 dollars at a fancy shop. How many dollars per bar?

9 :pæp

(!) Watch out: Keep the order straight — miles per hour is miles $\div$ hours. Dividing the other way gives hours per mile, a different (and usually tiny) number.

Using a Unit Rate: Multiply Back Up

Rule: Once you know the rate for **one** unit, any amount is one multiplication: total = unit rate $\times$ number of units.

Example: Stickers cost 9 cents each. How much do 6 stickers cost?

Step 1. The unit rate (9 cents per sticker) is given, so multiply by the number of stickers.

Step 2. $9 \times 6 = 54$, so 6 stickers cost **54** cents.

Try it: A ferry ticket costs 7 dollars per person. How many dollars do 8 people pay?

95 :pæp

Comparing Two Rates Fairly

Rule: To compare two deals or speeds with **different** sizes, bring both to a unit rate first, then compare the unit rates. Smaller is better for prices; bigger is better for speeds and distances.

Example: Bag P holds 4 muffins for 24 dollars; bag Q holds 6 muffins for 30 dollars. Which costs less per muffin?

Step 1. Unit prices: $24 \div 4 = 6$ dollars per muffin, and $30 \div 6 = 5$ dollars per muffin.

Step 2. Comparing them, bag Q is cheaper per muffin: $6 > 5$.

Try it: Runner X covers 18 km in 3 hours; runner Y covers 20 km in 4 hours. Compare their km-per-hour fractions with $<$ or $>$.

check: $\frac{4}{20} < \frac{3}{18}$

Key Vocabulary

Unit rate: the amount for exactly one unit

Per: “for each one”; names what you divide by

Unit price: cost of one item — a unit rate for money

Rate table: a ratio table with a column for one unit